

STAT 20000: Elementary Statistics
Practice Midterm
April 18, 2026

Name: _____

UChicago ID: _____

Question 1.

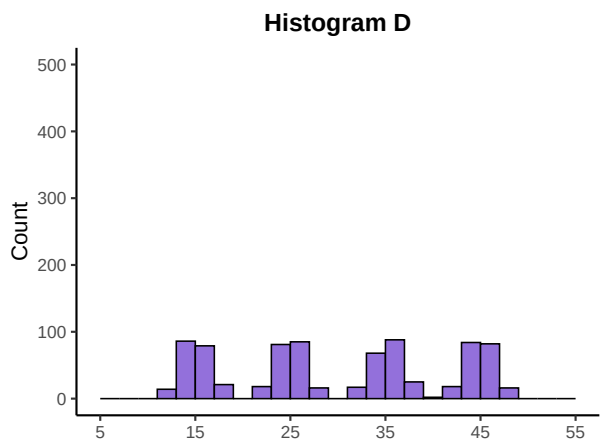
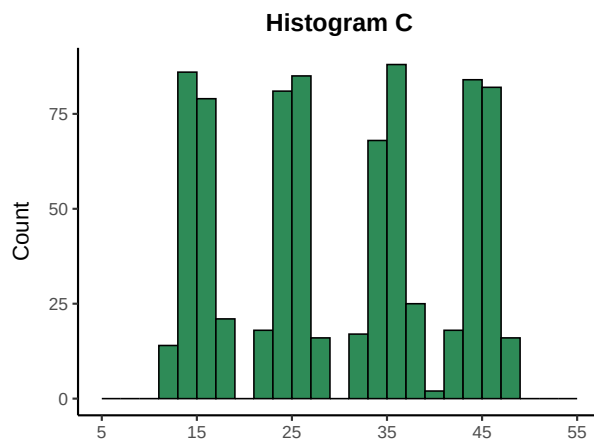
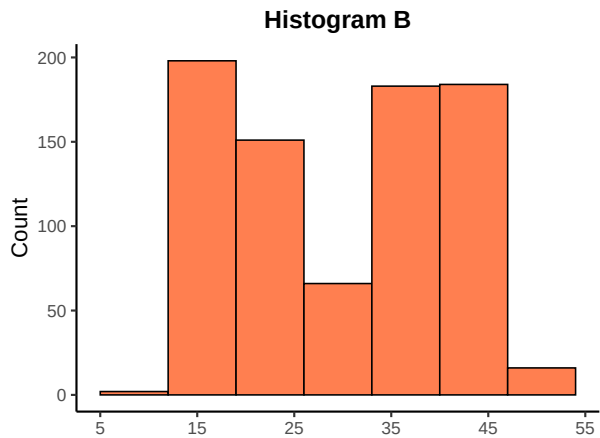
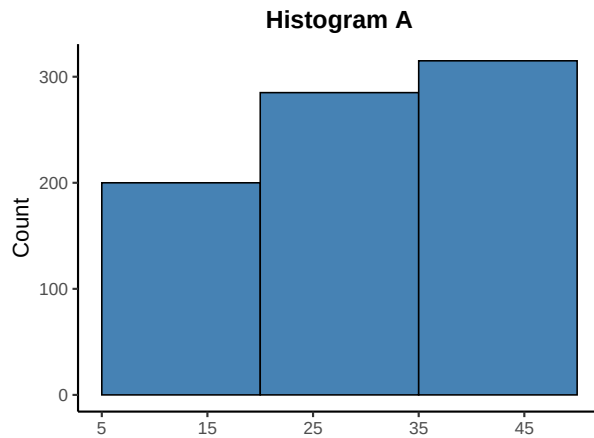
A university has two graduate departments: Engineering and Humanities. The admission rates for Group **A** and **B** applicants are shown below.

A	Applied	Admitted	Rate	B	Applied	Admitted	Rate
Engineering	400	240	0.600	Engineering	50	35	0.700
Humanities	50	15	0.300	Humanities	400	140	0.350
Total	450	255	0.567	Total	450	175	0.389

An advocacy group looks at the Total row and claims the university discriminates against **B** in admissions. Is their conclusion justified? Explain what is really happening in this data.

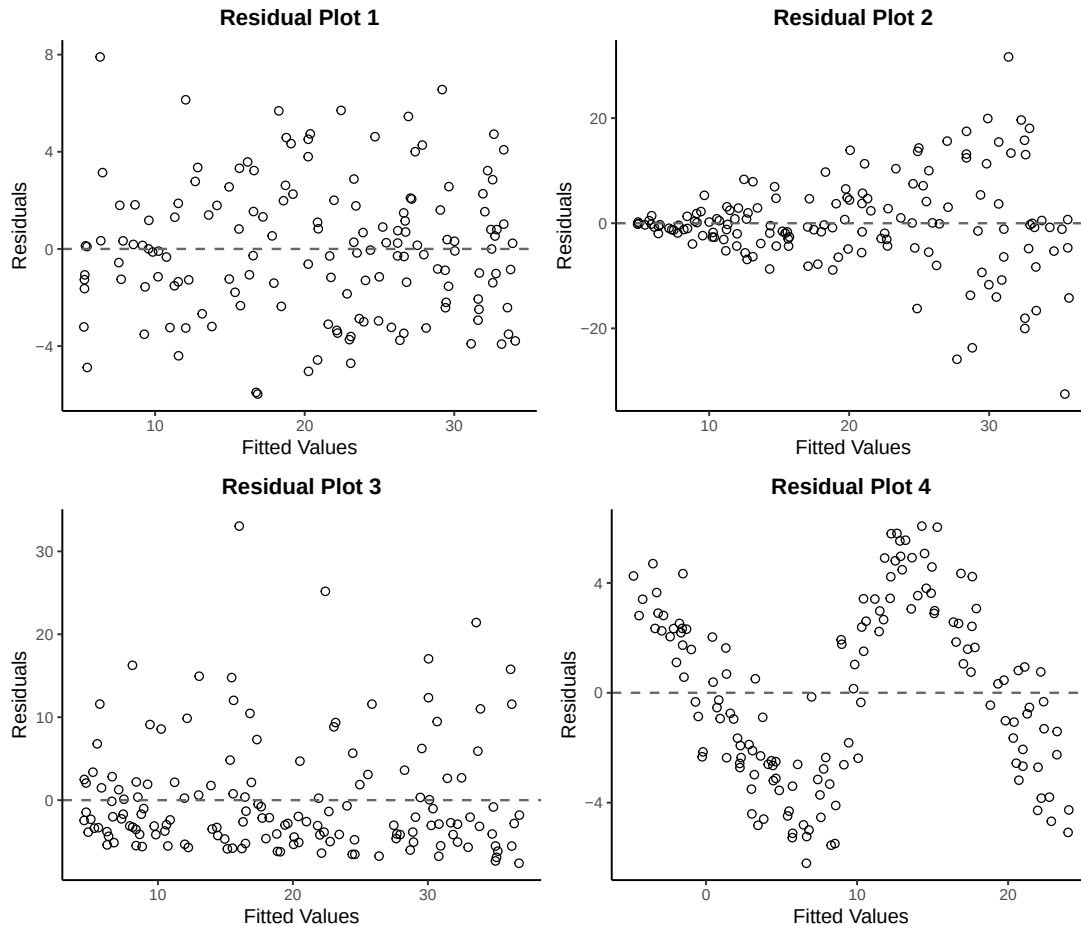
Question 2.

In the following we have histograms made by four analysts for the same dataset. Which histogram do you think presents the best visual representation of the data? Explain briefly.



Question 3.

In the following we have scatter plots of regression residuals for four data sets. For each case, decide whether a linear regression model is appropriate. If it is, briefly explain what supports its use. If it is not, identify what feature of the plot indicates that a linear regression is inadequate/inapplicable.



A study measured the *forearm length* (cm) and *height* (cm) of 120 adult males. Below are the data summary.

a. (5 pts) Predict the Height (in cm) of a man whose Forearm Length is 23 cm using the regression method. Show your calculation.

b. (3 pts) Find the root-mean-square error (r.m.s. error) for your prediction in part (a). Show your calculation.

c. (4pts) Write down the equation of the regression line for predicting Height from Forearm Length. Show your calculation.

- d. (4 pts) Predict the Forearm Length (in cm) of a man who is 190 cm tall, using the regression method. Show your calculation.
- e. (3 pts) For about 95% of men, your prediction in the previous part will be within _____ cm of the actual forearm length. Show your calculation.
- f. (3 pts) If forearm length were instead measured in **inches** ($1 \text{ cm} = 0.394 \text{ in}$),
- the average forearm length in inches is _____ ,
 - the SD of forearm length in inches is _____ , and
 - the correlation between forearm length (in inches) and height (in cm) is _____ .

Question 5.

Find the average, median, and standard deviation (SD) for the list below. Please show your work for full credit.

10, 15, 13, 12, 14, 15, 12

Average: _____ Median: _____ SD: _____

Question 6.

A researcher collects data on x (temperature in $^{\circ}\text{C}$) and y (ice cream sales in dollars) for 50 summer days and finds $r = 0.65$. For each scenario below, state the new correlation and briefly explain. No calculation is needed.

- a. She converts every temperature value from Celsius to Fahrenheit using $F = 1.8C + 32$. What is the new correlation between temperature (in $^{\circ}\text{F}$) and sales?

- b. Her colleague, for reasons unclear, multiplies every sales value by -1 (making them all negative). What is the new correlation?

- c. She adds 1000 to every temperature value and subtracts 500 from every sales value. What is the new correlation?

- d. She swaps the roles of x and y — that is, she now treats sales as the explanatory variable and temperature as the response. What is the new correlation?

Question 7.

A data analyst is studying three different datasets. For each one, a colleague claims: “Approximately 68% of the observations fall within one standard deviation of the mean.” Decide whether this claim is likely correct for each dataset and explain your reasoning.

- a. **Dataset A:** Final exam scores for a large introductory biology course. The histogram of scores is roughly bell-shaped and symmetric, centered around 72 with scores ranging from about 40 to 100.
- b. **Dataset B:** Annual household income in a large metropolitan area. The distribution is heavily right-skewed, with most households earning between \$30,000 and \$80,000 but a long tail stretching past \$500,000.
- c. **Dataset C:** The commute times (in minutes) of employees at a company where roughly half work at a downtown office and the other half at a suburban campus. The histogram shows two distinct peaks — one around 15 minutes and another around 45 minutes — with a dip in between.

Question 8.

Pemberton Elementary School has been struggling with low reading scores. The principal reads several studies and makes a series of claims. For each, say whether the claim is justified and explain.

Claim 1. The principal finds data showing that across 30 countries, the correlation between *average hours of television watched per day* and *average national reading score* is $r = -0.85$. She tells parents: “*If your child watches more TV, their reading score will drop sharply.*”

- a. (3 pts) Is the principal justified in applying the country-level correlation to individual children? What is this statistical pitfall called?

Claim 2. The principal surveys all 200 students and finds that children who eat breakfast daily score 12 points higher on reading tests than children who skip breakfast. She concludes: “*Eating breakfast causes higher reading scores.*”

- b. (3 pts) Is this an observational study or an experiment? Can the principal’s causal conclusion be justified? Name a plausible confounding variable.

- c. Describe an experiment the school could run to test whether breakfast actually improves reading scores. Be specific about random assignment.

Claim 3. The school librarian reports that the 200 students collectively checked out 8,600 books last year. The median number of books checked out per student is 28.

- d. The principal announces: “*On average, each student checked out 28 books.*” Is this correct? If not, what is the actual average?

- e. A parent asks: “How many books did a typical student check out?” Would you report the mean or the median? Explain briefly.

Question 9.

A health magazine reports the following finding: “In a study of 2,000 adults, those who took Vitamin D supplements daily for three years had 15% higher bone density on average than those who did not.”

a. Is this an observational study or an experiment?

- ☐ Observational study ☐ Experiment ☐ Cannot tell from the description

b. What is the explanatory variable in this study?

- ☐ Bone density ☐ Whether the adult took Vitamin D supplements ☐ Age of the adult

c. Can we conclude that Vitamin D supplements *cause* higher bone density? Explain. If not, name a confounding variable and describe how it could explain the observed difference. Then describe how you would redesign this as an experiment.